07C — Deep-Dive Feature Audit: Sect Governance & Internal Ecosystem (Part 3 of 5)

Sect Architecture, Dormitory Bed Synchronization, Virtual Specialist Facilities, Feng Shui 10-Element Matrix, Leyline Grid, 5-Element Barrier Dome, Totem Pillars & Sacred/Infernal Shrines

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Core Game Philosophy**: **100% Emergent, Autonomous Xianxia Sandbox**. Every hall, dormitory, formation pillar, barrier dome, totem, and altar must exist coherently on the sect compound, scale its physical stations with building upgrades, respond autonomously to war/peace via the sect's `arraymaster` and `steward`, and generate authentic ideological friction when demonic blood altars collide with righteous disciples!
- **Master Roadmap for Category 07 (`07A–07E`)**:
  - `07A`: Leadership, 6 Directorates & 25 Official Posts, Jurisdictions, Stipends & Censor Audits
  - `07B`: The 8 Sacred Peaks (`SECT.PEAKS`), 144 Peak Campus Stations, Peak Vaults & Peak-Lord Autonomy
  - **`07C` (This Document)**: **Sect Buildings, Dormitory Housing Caps, Specialist Facilities, Feng Shui (`FENG_SHUI`), Leyline Grid (`LEYLINE_GRID`), Elemental Barrier (`ELEMENTAL_BARRIER`), Ancestral Shrine, Totems (`TOTEMS_BLOODLINES`) & Cosmic/Infernal Shrines**
  - `07D`: Internal Politics, 4 Elder Cliques, Standing Edicts, 5 Sect Charters, Strategic Council, Tribunal, Dungeon, Succession & Mutiny
  - `07E`: Grand Finale — Holistic Overview of the Entire Living Sect Ecosystem

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0. Hard Architectural Invariants

1. **NEVER modify `js/human_core.js`** (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **Preserve `v214`, `v320`, `v330` & `v410` Contracts**: Keep `Render.resolveBarrierColor(W)` hex codes (`METAL..EARTH`), `npc.sectTotems` save/load persistence (`v320` Phase 10), and `LEYLINE_GRID` anchor IDs intact.

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1. [BLD-DORM-01] Critical Housing Cap vs Physical Bed Station Desync: Up to 44 Disciples Become Homeless (n.bed === null) in a 100-Member Sect!

Compare SECT.housingCapacity(W) (gameplay-architecture Sec 8.1) against the actual 'bed' stations spawned in MAP.STN (js/world_map.js:76–78, 100–101):

1. **Formula vs Physical Reality Mismatch**:
   - SECT.housingCapacity(W) computes theoretical bed limits up to **100 members** (56 dorm_outer Lv.5 + 28 dorm_inner Lv.4 + 16 dorm_core Lv.3), while completely ignoring elder_quarters (10 beds) and founded Peak Sanctuaries (6 beds each = up to 48 beds).
   - Meanwhile, MAP.create (world_map.js:100–101) initializes all 3 dormitories at level: 1 (lines 76–78) with **24 + 14 + 8 + 10 = 56 static 'bed' stations**!
   - **Catastrophic Late-Game Consequence**: If SIM.finishBuilding / expandStations does not dynamically append new physical 'bed' stations into W.map.stations when dorm_outer, dorm_inner, and dorm_core upgrade from Lv.1 $\to$ Lv.5, **recruiting members 57..100 leaves up to 44 disciples permanently homeless (n.bed === null)**, unable to rest properly or fall back to bed meditation!
2. **Rank Promotion Never Upgrades Dormitory Tier**:
   - When an Outer Disciple (rank 1, sleeping in dorm_outer) is promoted to Core Disciple (rank 3) or Elder (rank 4), n.bed stays pointing to their cramped dorm_outer bunk forever while elder_quarters and dorm_core beds sit empty (blocking new novices from taking dorm_outer beds)!
- **Exact Fix**:
   - Make SECT.housingCapacity(W) equal the **exact count of active physical 'bed' stations** across dorm_*, elder_quarters, and **founded** Peak Sanctuaries (s.active !== false), dynamically adding +6..+8 indoor virtual bed slots (where === 'in', zero map-tile collision!) whenever a dormitory upgrades!
   - Re-evaluate AI.assignBed(W, n) on rank promotion (moving Rank 4+ to elder_quarters, Rank 3 to dorm_core, Peak members to their founded Peak Sanctuary).

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2. [BLD-VIRT-02] Ghost Specialist Facilities (mech_workshop, beast_menagerie, harmonic_pavilion) Lack Physical Map Presence & Never Upgrade Above Lv.1!

- **Locations**: `js/sect.js` (`W.sect.specialistFacilities`), `js/world_map.js:24-115` (`MAP.PLOTS`)
- **Two Structural Gaps**:
  1. **Zero Physical Map Presence**: `mech_workshop`, `beast_menagerie`, and `harmonic_pavilion` exist purely as numbers `{ level: 1 }` on `W.sect.specialistFacilities`—they are absent from `W.map.byId` (breaking `POST_CATALOG.beastmaster.hall` and `steward.hall` in `07A`!), while the 4 reserved expansion plots `MAP.PLOTS` (`plot_w`, `plot_e`, `plot_n`, `plot_se` in `world_map.js:110-115`) sit unused forever!
  2. **Zero Autonomous Upgrades (`Stuck at Lv.1 Forever`)**: `SECT.upgradeBuilding(W, master)` scans `W.map.buildings` via `facilityDemand`, never upgrading `W.sect.specialistFacilities`!
- **Fix**: Bind `mech_workshop` $\leftrightarrow$ `peak_array`/`forge`, `beast_menagerie` $\leftrightarrow$ `peak_beast` (or instantiate them on `MAP.PLOTS`), and teach `SECT.masterDecision` / `steward` to upgrade `specialistFacilities` (`Lv.1 -> Lv.5`) whenever the sect has ≥ 3 construct/beast/music cultivators and surplus Spirit Stones!

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3. [BLD-TOTEM-03] Newly Recruited Disciples Receive 0% Bonus From Existing Sect Totem Pillars (SECT.TOTEMS_BLOODLINES in sect.js & v320:240)!

- **Exact Proof (`test/v320_dormant_awakening_suite.js:240-248` & `js/stats.js:331, 560`)**:
  - Look at how `PROP-33` persisted Sect Totems so `getUnitBonuses(n)` (`1` argument, no `W`!) survived Save/Load: `erectTotemPillar(W, totemId)` stamps `totemId` into **`npc.sectTotems` on NPCs currently alive at that instant**!
  - **Bug**: Every disciple recruited, born, or defecting into the sect **after** `erectTotemPillar` was called (`SIM.addNpc`, `acceptApplicant`, `RECRUIT_DEFECTOR`) starts with `npc.sectTotems === undefined`—**receiving `0%` bonus from the Sect's own standing Totem Pillars for their entire life**!

- **Exact Fix**: Store authoritative totems in `W.sect.totems` **and** automatically sync `npc.sectTotems = W.sect.totems.slice()` inside `SIM.addNpc`, `SIM.dailyUpdate`, and `SIM.deserialize` (preserving 100% compatibility with `v320` Phase 10's `G.W = null` browser test!), plus let `beastmaster` / `BEAST_SOVEREIGN` / `GOD.TOTEM_BEAST_GOD` elders autonomously erect totem pillars when `W.sect.totems.length < maxTotems`!

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4. `[BLD-ARRAY-04]` Five Overlapping Formation/Barrier Systems With `5`-vs-`10` Element Mismatch, Dead-Guard Exploit & `0%` `arraymaster` Autonomy!

`JS2-v2.6.0` contains **five separate sect-defense/formation layers** (`SECT.FENG_SHUI`, `SECT.LEYLINE_GRID`, `SECT.ELEMENTAL_BARRIER`, `CULT.MANTRA_SYSTEM.setSectFormationMode`, and `anchorDragonVeinNode`):

1. **`5-Element` vs `10-Element` Vocabulary Mismatch on `ELEMENTAL_BARRIER` (`v410:191` vs `combat.js`)**:
   - `ELEMENTAL_BARRIER.setBarrierElement` uses `'METAL'`, `'WOOD'`, `'WATER'`, `'FIRE'`, `'EARTH'`, whereas combat skills and raiders use **`'EARTH_METAL'`**, **`'WOOD_LIFE'`**, **`'WATER_ICE'`**, **`'FIRE'`**, **`'WIND'`**, **`'LIGHTNING'`**, **`'LIGHT_YANG'`**, **`'YIN_SHADOW'`**, **`'ASTRAL'`**, **`'BLOOD_DEMON'`**!
   - Normalize element matching via `ELEM` (`METAL <-> EARTH_METAL/LIGHTNING`, `WOOD <-> WOOD_LIFE/WIND`, `WATER <-> WATER_ICE/YIN_SHADOW`, `FIRE <-> FIRE/LIGHT_YANG`, `EARTH <-> EARTH_METAL/ASTRAL`) so the 5-Element Dome genuinely counters incoming elemental raids!
2. **Corpse / Absent Guard Exploit on `LEYLINE_GRID` (`v330:157`)**:
   - In `SECT.LEYLINE_GRID.tickContestation(W)`, verify that `guard = W.byId[anchor.guardNpcId]` is **alive, not exiled, not detained, and `!guard.away`** before granting `guarded: true` (`12` reduced damage & `+10` `merit`)! Auto-reassign a living `guard`/`vanguard`/`arraymaster` disciple when an anchor guard dies or leaves!
3. **Zero Autonomous `arraymaster` / Geomancer Control**:
   - Wire `SECT.runOfficerJurisdictionSingle(W, 'arraymaster')` (and raid-start hooks in `EXPED`) so a skilled `arraymaster`:
     - Switches `setSectFormationMode` to **`GRAND_GUARDIAN_TURTLE`** / **`SLAUGHTER_SWORD_MATRIX`** when `W.threat.raid` or `W.threat.alert` triggers, and restores

SPIRIT_GATHERING_CONVERGENCE (+18% meditation speed) once peace returns!
- Tunes ELEMENTAL_BARRIER.setBarrierElement to counter the raiding rival sect's primary element!
- Progressively attunes clashing buildings in SECT.FENG_SHUI.attuneBuilding and assigns guards to unguarded LEYLINE_GRID anchors!

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5. [BLD-SHRINE-05] Sacred vs Infernal Shrine Moral Clash, Dormant COSMIC_SHRINE (N-03), & Unrepaired Building HP!

1. **INFERNAL_SHRINE Buffs Righteous Monks With Zero Moral Friction (stats.js:354)**:
   - Consecrating the Blood-Sacrifice INFERNAL_SHRINE grants global +atkBonusPct to every NPC while righteous BUDDHA_DHARMA / high-integrity disciples feel **zero** disgust! Make orthodox disciples (righteousness >= 65 or BUDDHA_DHARMA) lose sat/loyalty (and refuse blood-altar buffs, preferring ancestral / COSMIC_SHRINE blessings) if a Demonic Master erects a Blood Altar, while STRICT_ORTHODOX_BAN seals INFERNAL_SHRINE!
2. **Dormant SECT.COSMIC_SHRINE.consecrateOffering (N-03 in v2.3.0 Audit)**:
   - Currently called **only** by v220 tests (0 live calls)! Wire A.pray at shrine for DIVINE_SUMMONER, DIVINE_POSSESSION, DIVINE_CHANNELER, and GOD elders to perform consecrateOffering organically!
3. **Ancestral Shrine Tablet Integration & Building Structural Repair**:
   - Connect W.sect.legacyTablets count directly to ANCESTRAL_RITUAL power scaling (01G) and ensure steward / construction repairs any b.hp < 100 damaged during seasonal calamities (CALAMITIES_4_SEASONS) or sieges!

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6. Mandatory Verification Suite Additions (07C)

1. **100-Member Physical Bed Invariant Test**: Upgrade dormitories to max level, spawn 100 NPCs, run AI.assignBed, and assert **0** living non-exiled disciples have n.bed === null.
2. **Post-Erection Totem Inheritance Test**: Call erectTotemPillar(W, 'TOTEM_WHITE_TIGER'), then add a brand-new disciple via SIM.addNpc, and assert TOTEMS_BLOODLINES.getUnitBonuses(newDisciple).active === true.

3. **Autonomous `arraymaster` War/Peace Formation Switch Test**: Trigger a raid with an appointed `arraymaster` and assert formation posture shifts to defense/slaughter, then returns to `SPIRIT_GATHERING_CONVERGENCE` after raid resolution.